Statistical analysis for musical corpora

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Plan

- Families of data modelling approaches
- Data representations and feature extraction
- Data visualization
- Statistical tests

Do basses have it the worst?

- In SATB choirs, sopranos usually sing the melody, while ATB voices do harmony
- The ATB parts are more difficult to sing, but do basses have the hardest part?

https://gertim-alberda.com/chorales/BachChorales/B004.html

Workflow

Define the question that you are investigating (state your hypothesis)

Find the data and clean, curate, prepare the dataset

Find a representation of your data compatible with your hypothesis

Do statistical Modeling (or ML), Build a model

Evaluate your model / results

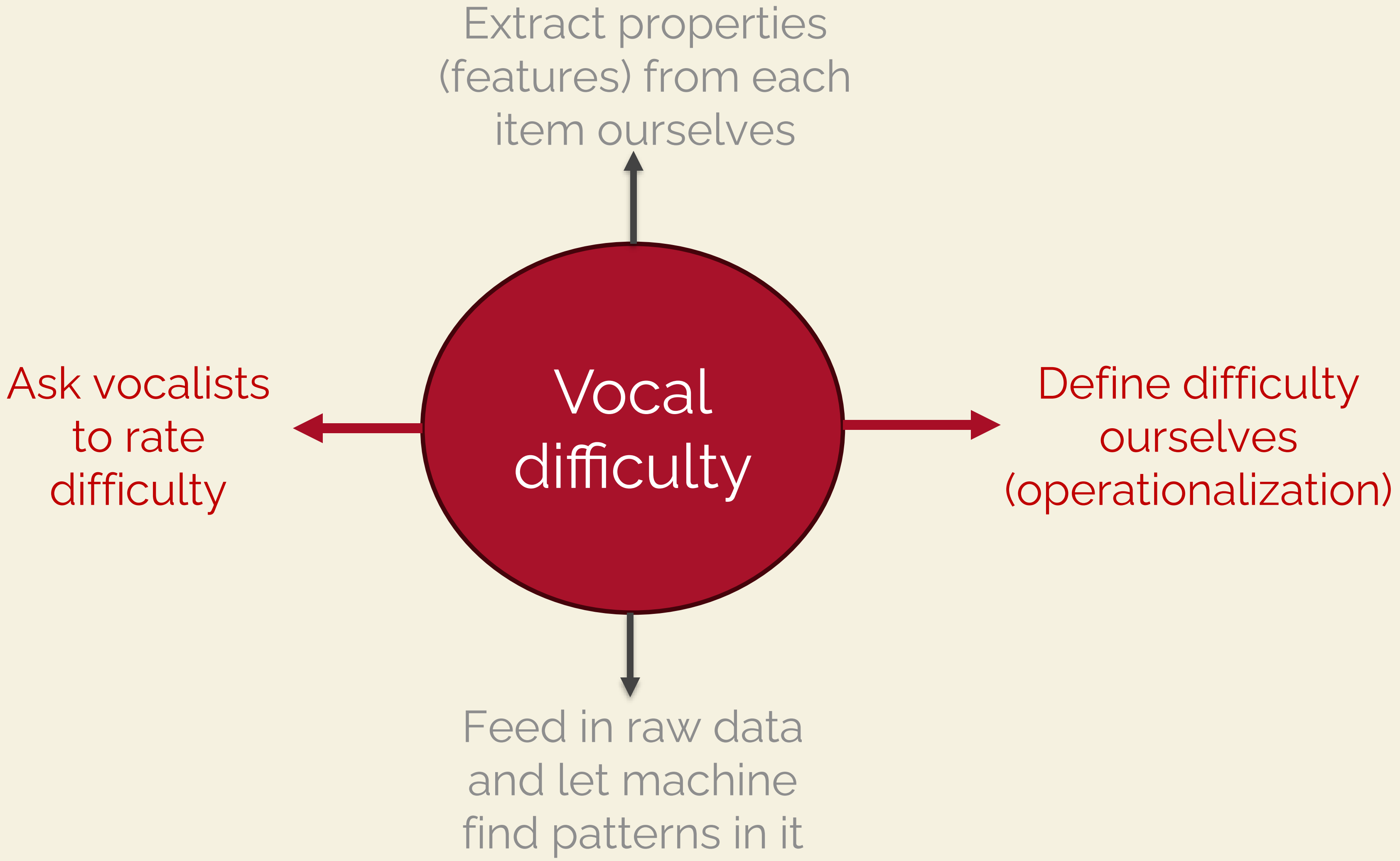
Interpret results

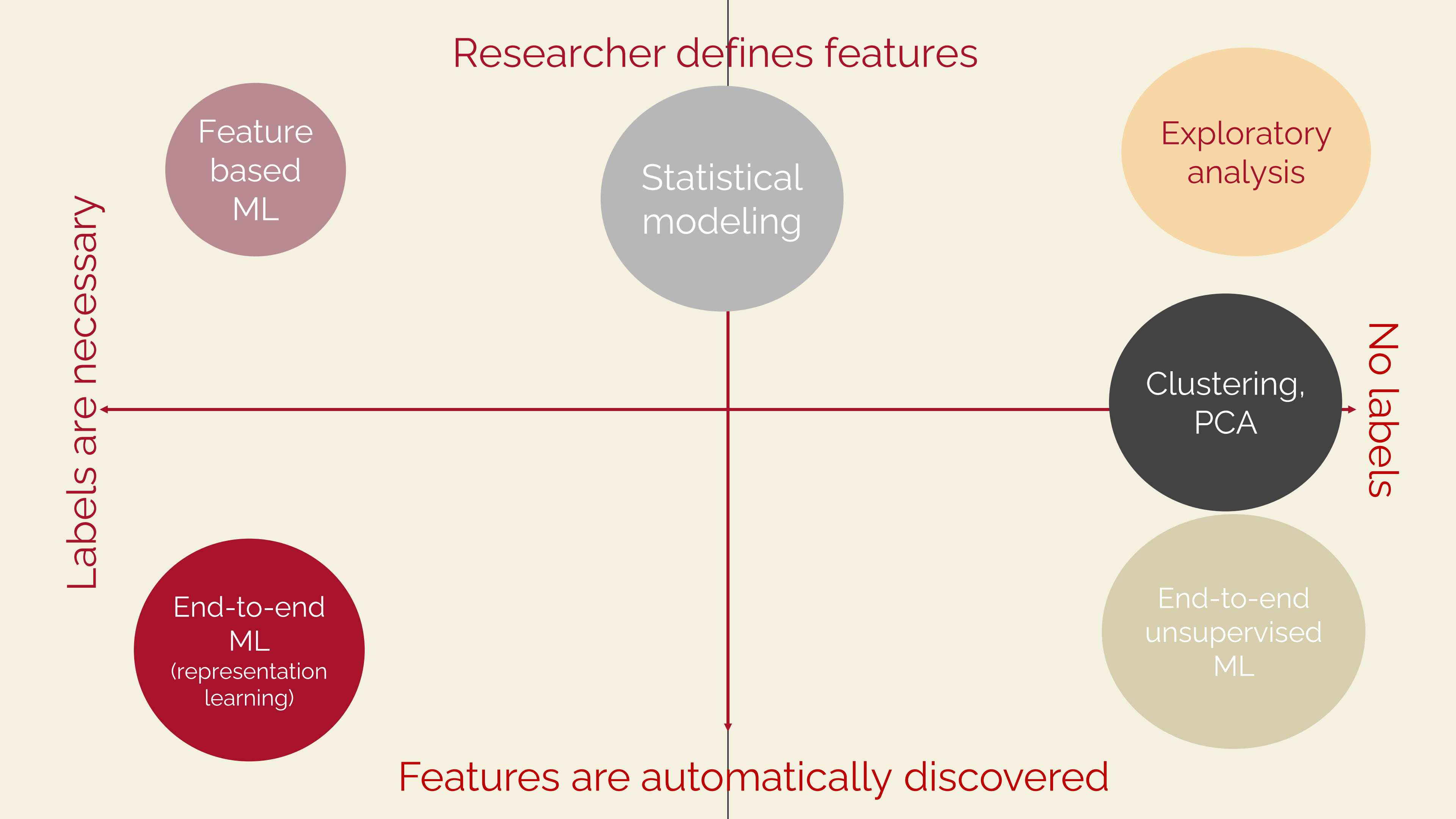
Our research project

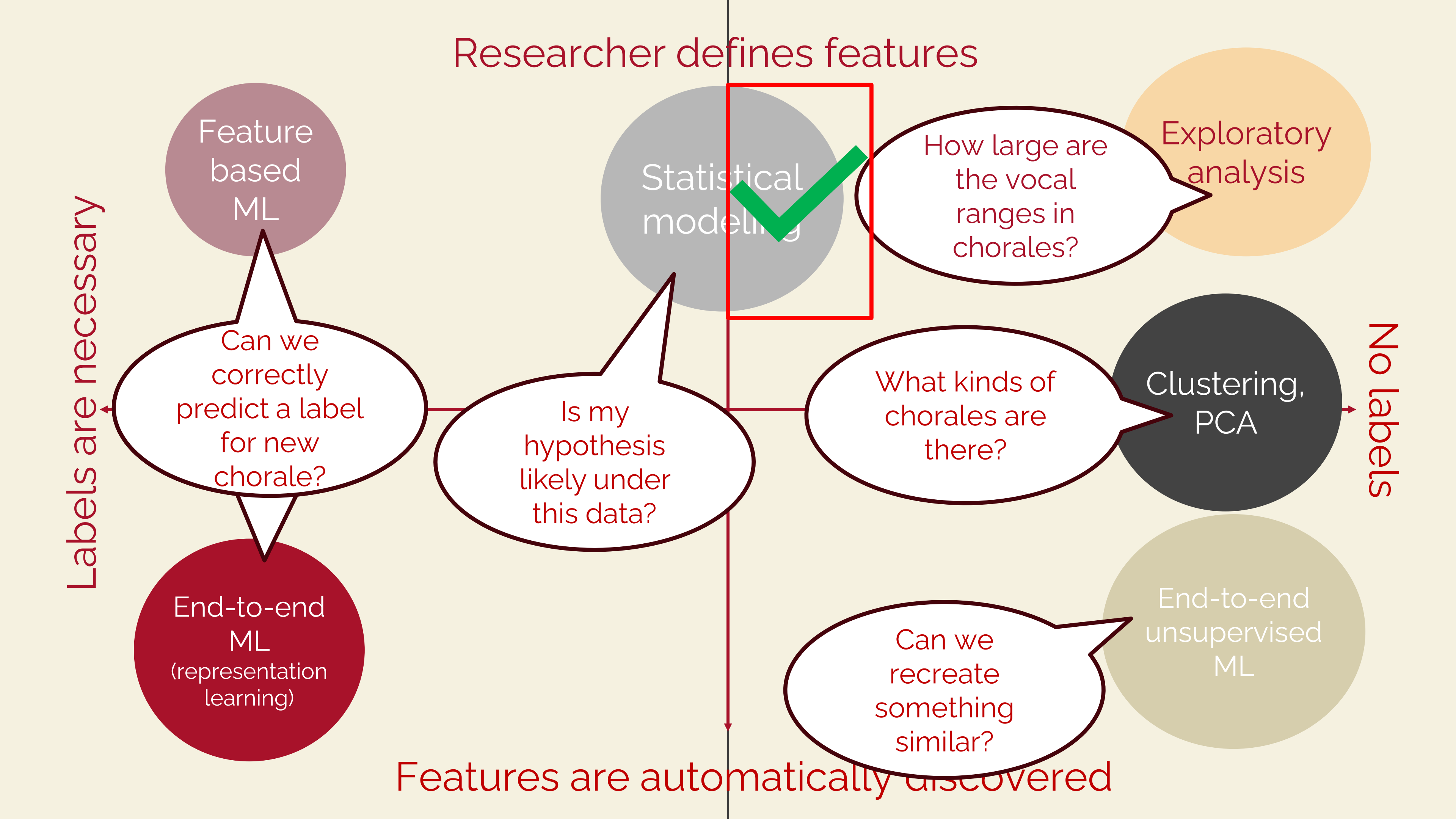
Bass parts in Bach chorales are the most difficult to sing.

A corpus of 320 Bach chorales from music21

Find a representation of our data compatible with our hypothesis


Extract properties (features) from each item ourselves
Ask vocalists to rate difficulty
Vocal difficulty
Define difficulty ourselves (operationalization)
Feed in raw data and let machine find patterns in it


Researcher defines features
Features are automatically discovered
Labels are necessary
No labels
Feature based ML
Statistical modeling
Exploratory analysis
Clustering, PCA
End-to-end ML (representation learning)
End-to-end unsupervised ML


Researcher defines features
Feature based ML
Statistical modeling
Exploratory analysis
How large are the vocal ranges in chorales?
Labels are necessary
Can we correctly predict a label for new chorale?
Is my hypothesis likely under this data?
What kinds of chorales are there?
Clustering, PCA
No labels
End-to-end ML (representation learning)
Can we recreate something similar?
End-to-end unsupervised ML
Features are automatically discovered

What influences singing difficulty – according to you?

Vocal difficulty

- Jumps larger than 4 semitones
- Accidentals
- Tessitura extremes
- Syncopation
- Long phrases
- Phonematic difficulty
- Fast tempo
- Ornamentation

Our research project

Bass parts in Bach chorales are the most difficult to sing.

A corpus of 320 Bach chorales from music21

Large jumps (%), accidentals (%), tessitura extremes

Statistical tests (t-test, Chi squared test, ANOVA)

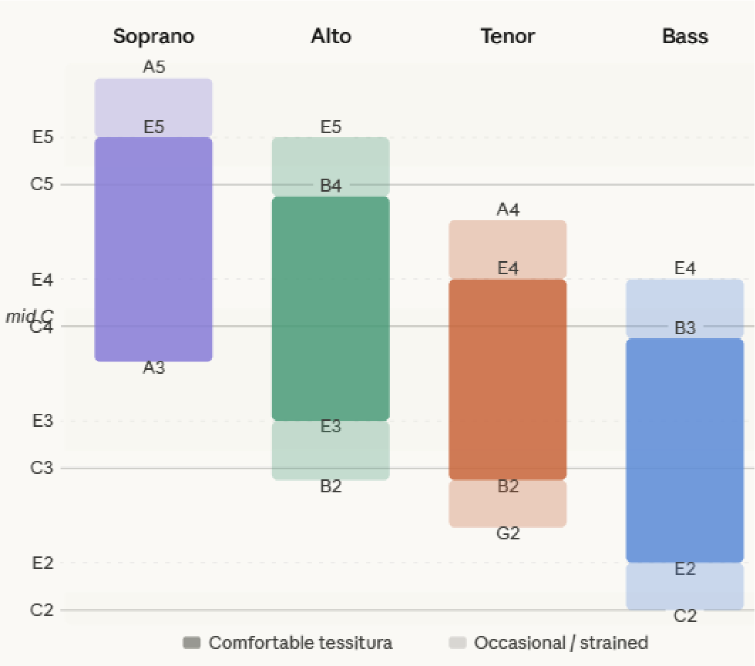
Are the result statistically significant?

Basses do or do not have the short end of the stick

Descriptive statistics

Summarizing musical data with numbers

Extracting features



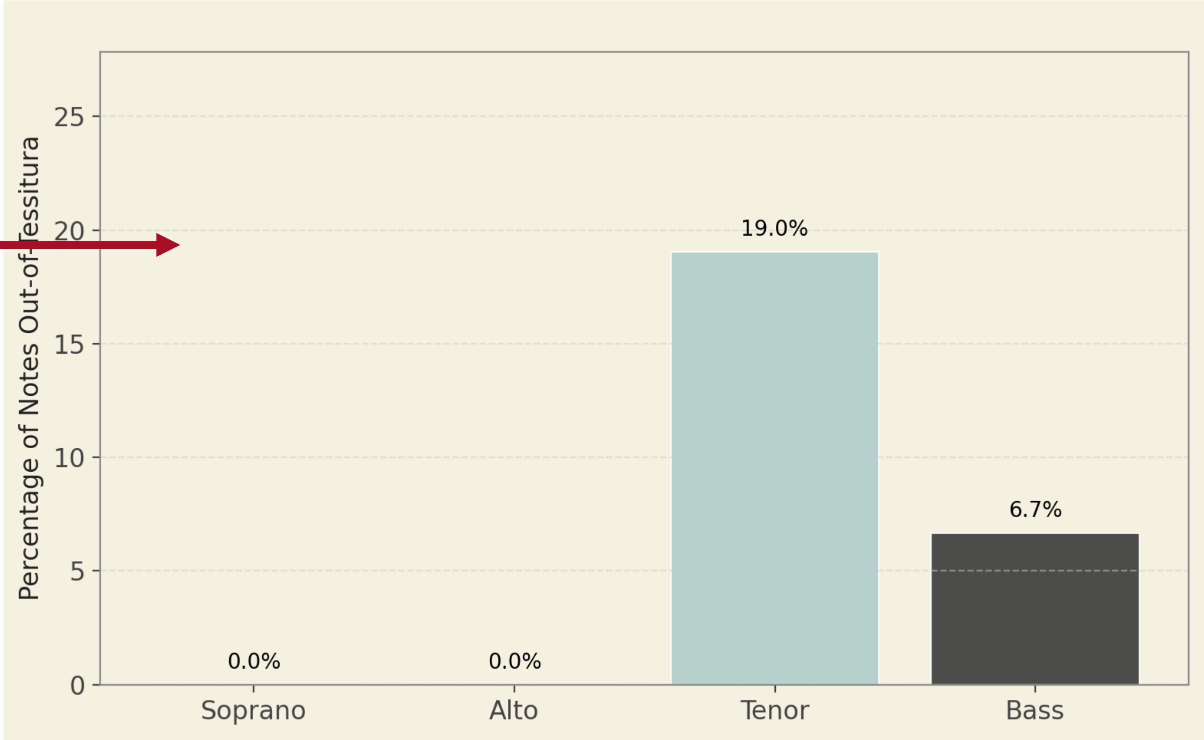

```python
def compute_tessitura_stats(pitches, part_name):
    tessitura = {
        "Soprano": lambda p: 57 <= p <= 76,
        "Alto":    lambda p: 53 <= p <= 72,
        "Tenor":   lambda p: 47 <= p <= 64,
        "Bass":    lambda p: 40 <= p <= 59
    }

    in_range = tessitura[part_name]
    out_count = sum(not in_range(p) for p in pitches)

    return 100 * out_count / len(pitches)
```

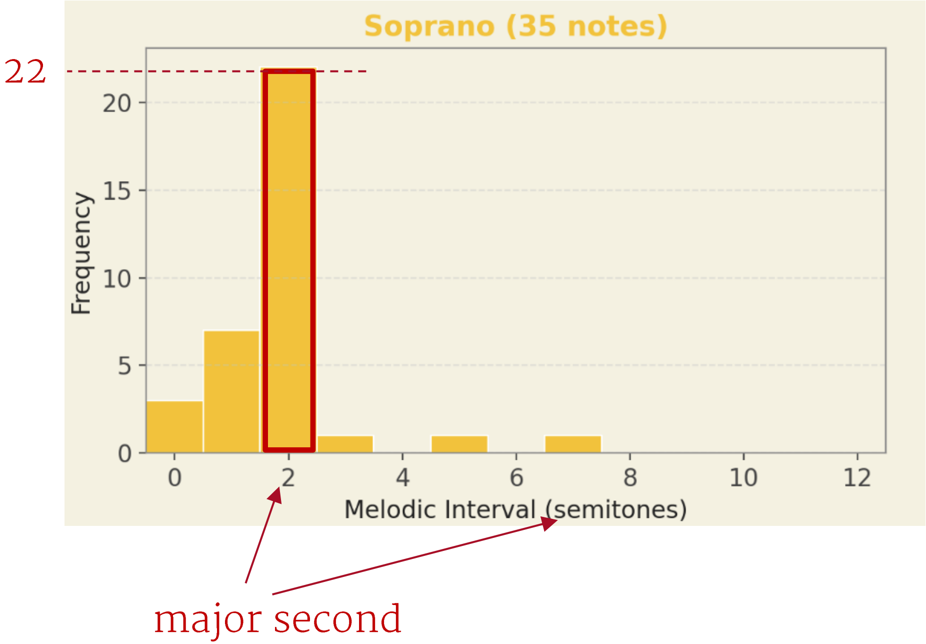
Chorale BWV 43.8: Ach Gott und Herr

Bar plot



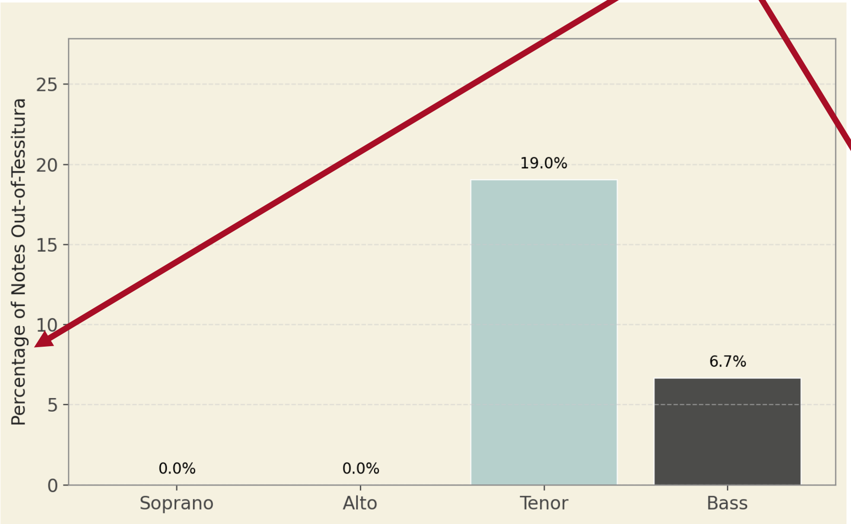
Data Distribution

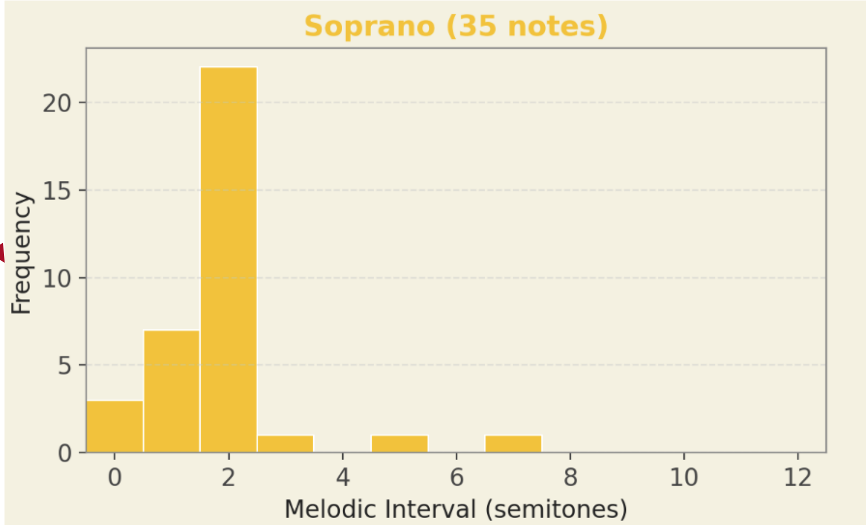
- A distribution shows how often different values occur in our data
- A distribution can be shown on a **histogram**



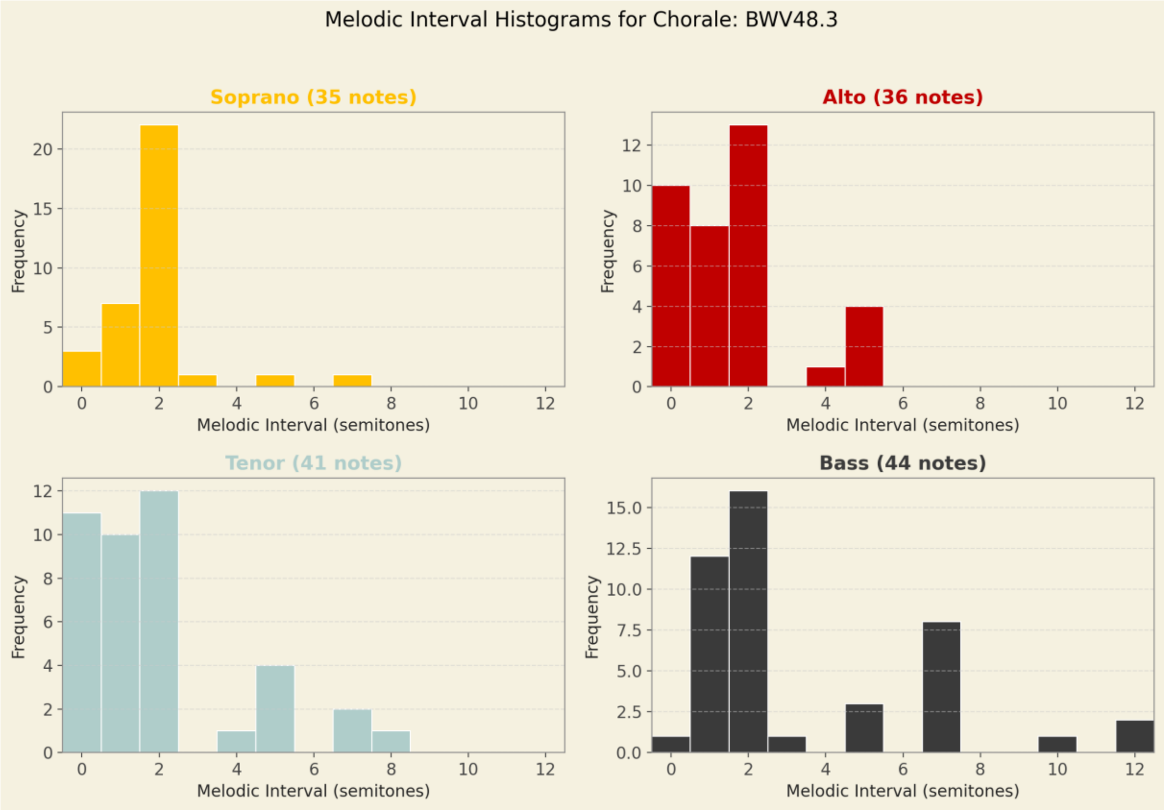
Absolute vs relative frequency

Actual count or percentage?





Data Distribution



The Mean (μ)

Concept

- The mean is the sum of all values divided by the number of values
- Formula: $\mu = (x_1 + x_2 + \ldots + x_n) / n$
- Sensitive to extreme values (outliers)
- Best used when data is roughly symmetrical

Musicological example

- Musical example: mean melodic interval in soprano voice
- If soprano has intervals [2, 1, 4, 2, 7, 1, 2] semitones...
- Mean = $(2+1+4+2+7+1+2)/7 = 2.71$ semitones
- One large leap (7) pulls the mean upward

The Median

Concept

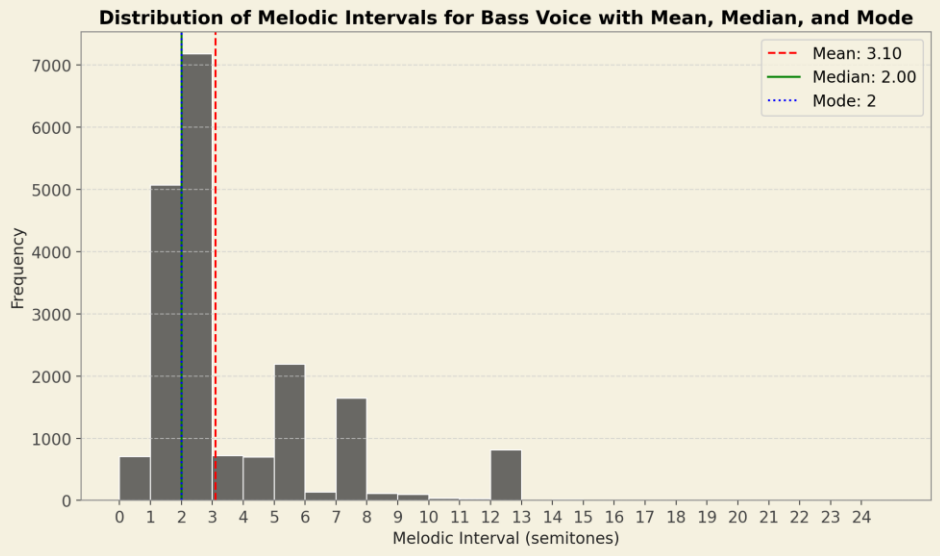
- The median is the middle value when data is sorted
- 50% of values fall below, 50% above
- Robust to outliers — not affected by extreme values
- Preferred when data is skewed

Musicological example

- Same soprano intervals sorted: [1, 1, 2, 2, 2, 4, 7]
- Median = 2 semitones (the middle value)
- More representative than mean when leaps are rare but large
- Useful for comparing 'typical' difficulty across voices

Skewness: shape of the distribution

- Oftentimes a distribution has a long tail in one direction (is skewed)
- In a right-skewed distribution median < mean
- For example, here a “typical” interval is smaller than average interval.



The Mode

Concept

- The mode is the value that occurs **most frequently** in the distribution
- This is the only statistic which might be not unique – there can be several values that are a mode

Musicological example

- Soprano intervals [1, 1, 1, 2, 2, 2, 4, 7]
- Mode: 1 and 2

Example

A melody consists of the following MIDI pitches:

51, 57, 48, 53, 52, 54, 51

sorting

↓

48, 51, 51, 52, 53, 54, 57

What is the mode of this distribution?

What is the median?

Standard Deviation (σ)

Standard deviation (SD) measures how spread out values are around the mean.

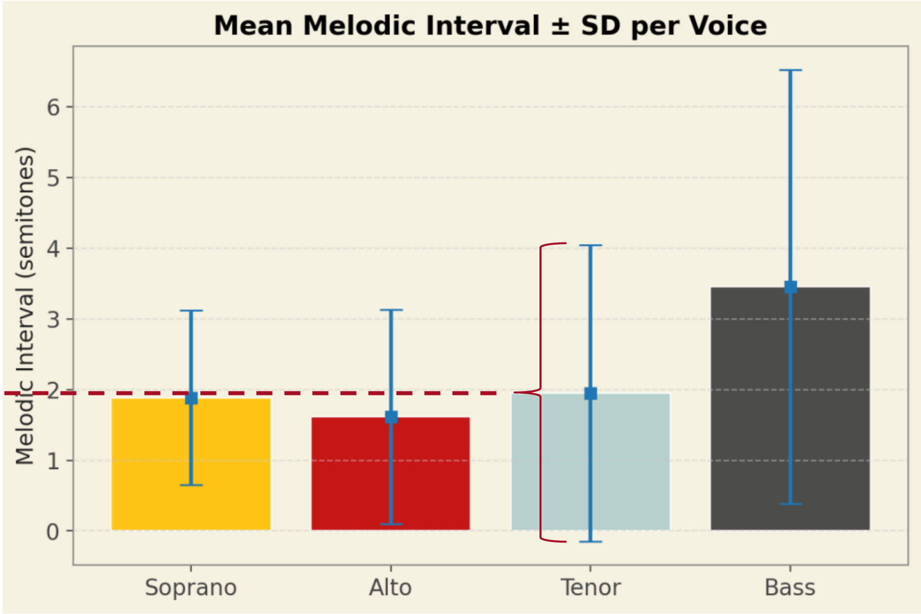
- Small SD → values cluster near the mean
- Large SD → values are widely scattered

Formula: $\sigma = \sqrt{[\, \Sigma(x_i - \mu)^2 / n \,]}$

<u>Musical interpretation:</u>

A high SD in melodic intervals means a voice has very varied leaps - sometimes stepwise, sometimes large jumps.

SD ≈ 4



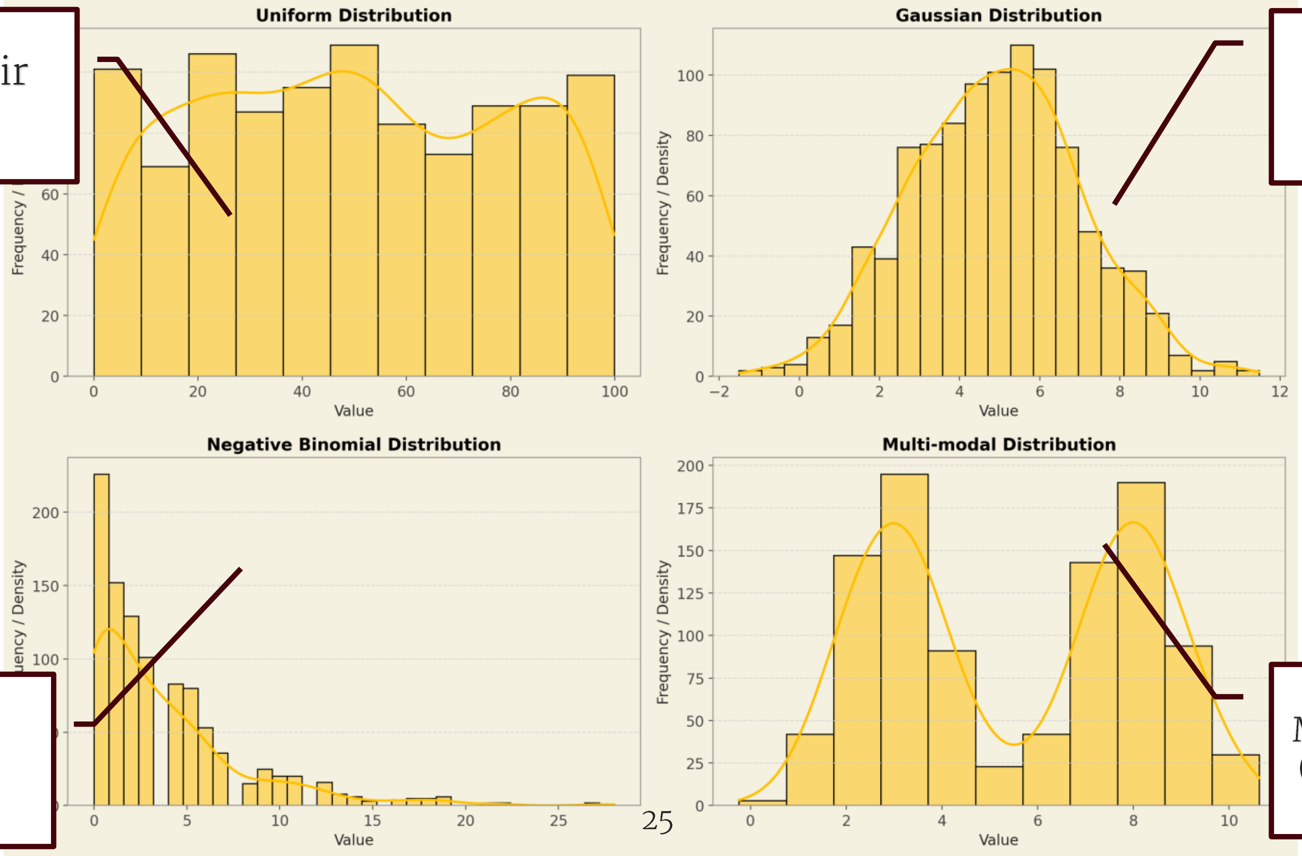
Types of data distributions

- Data takes different shapes depending on the underlying real-world processes:
  - Constraints (maximum/minimum MIDI pitch on a piano)
  - Mixture of different groups (beginners/experts)
  - Natural variation

Types of data distributions

Rolling a fair die

Human height
IQ scores

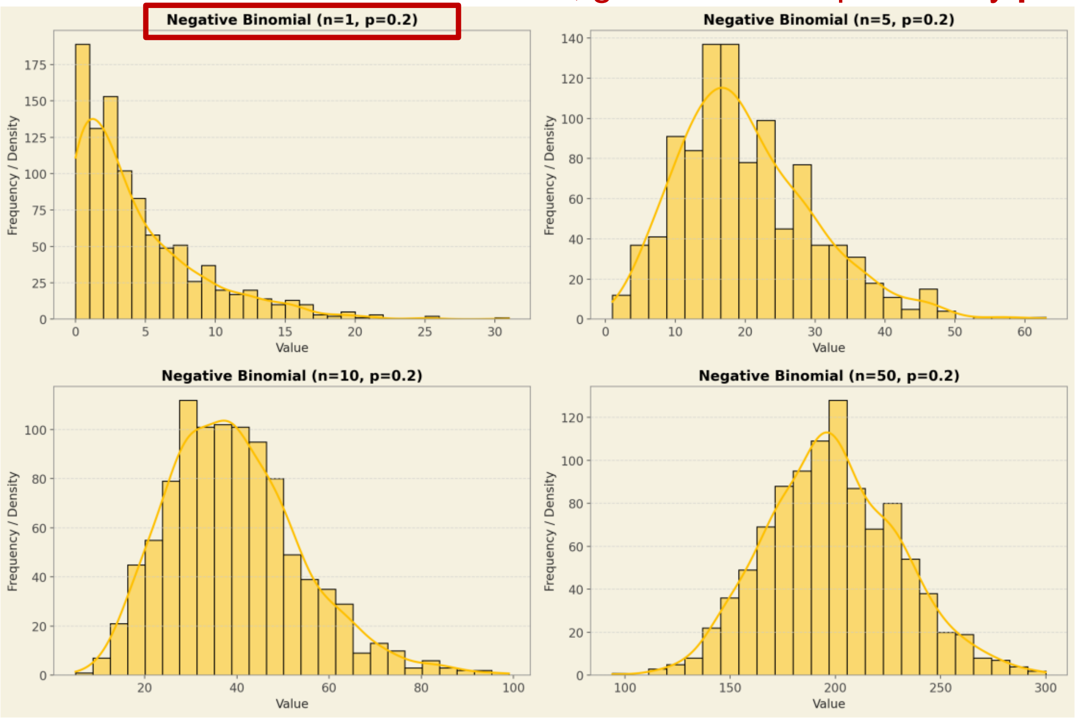


Music streaming counts

Mixed populations (different genres)

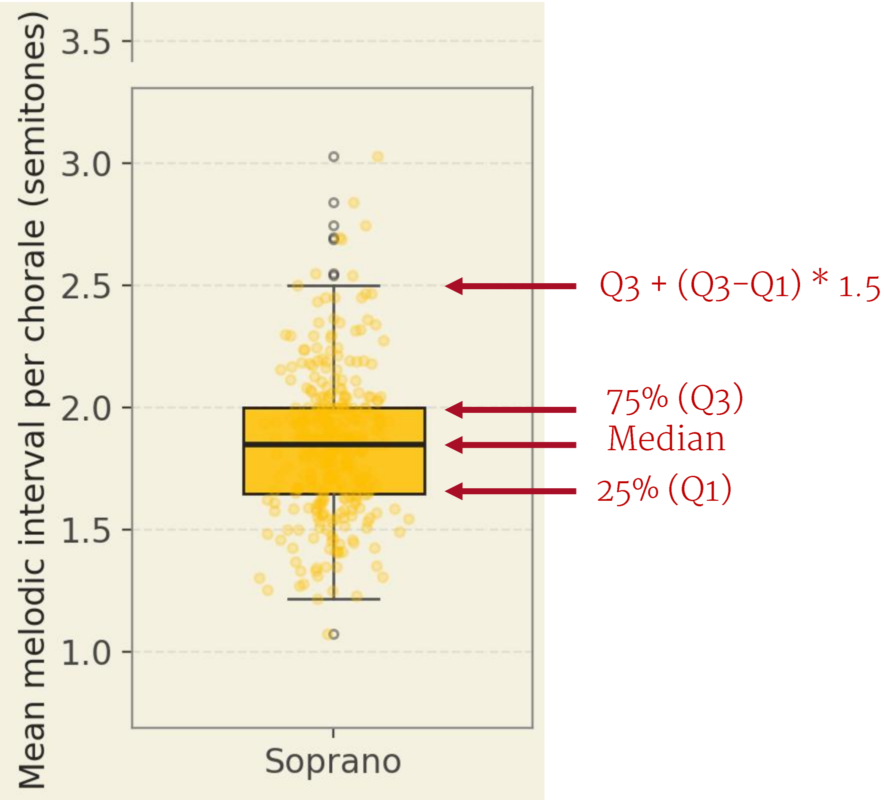
Distribution parameters

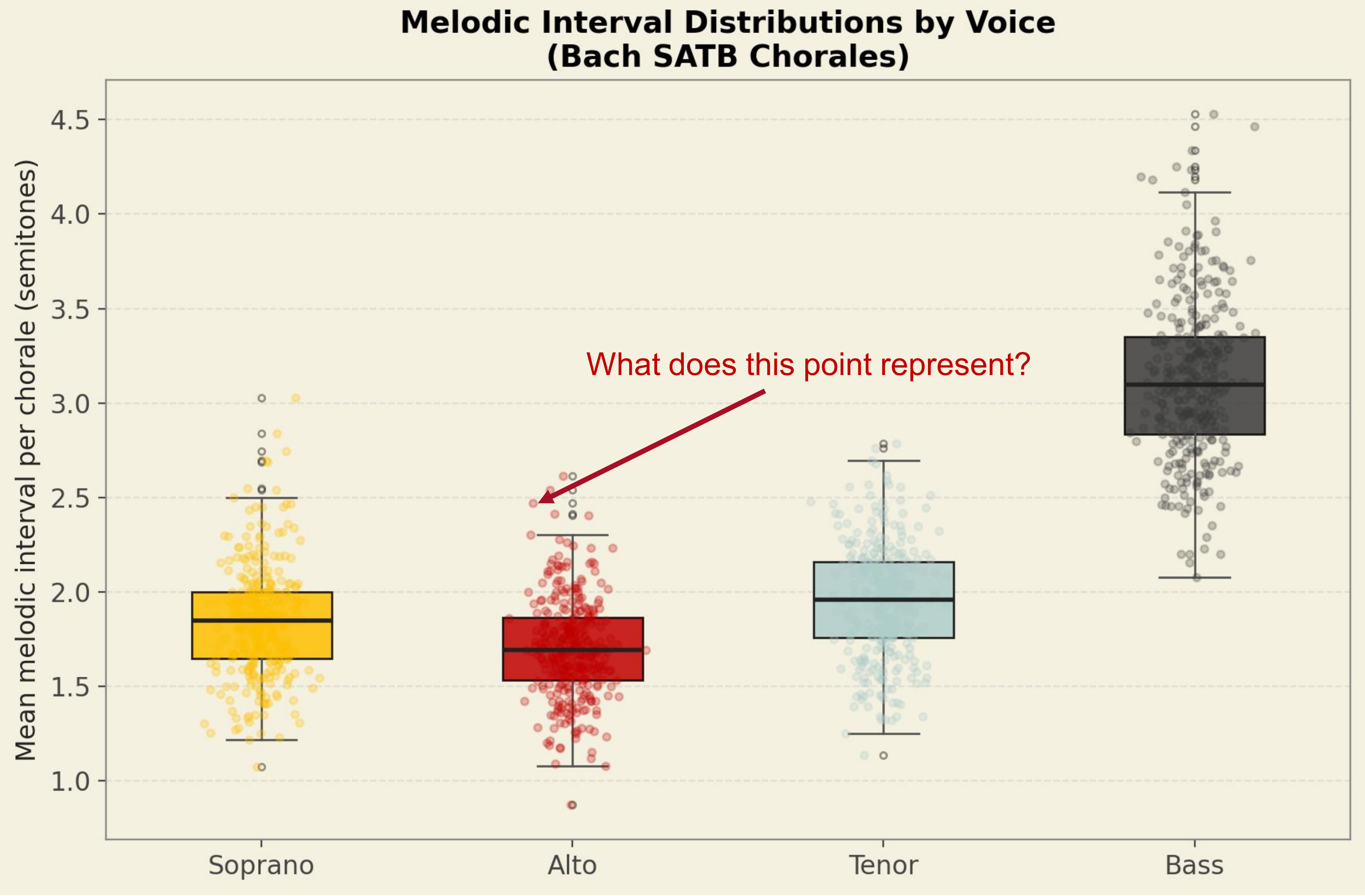
How many failures occur before achieving **n=1 successes**, given success probability **p=0.2** per trial?

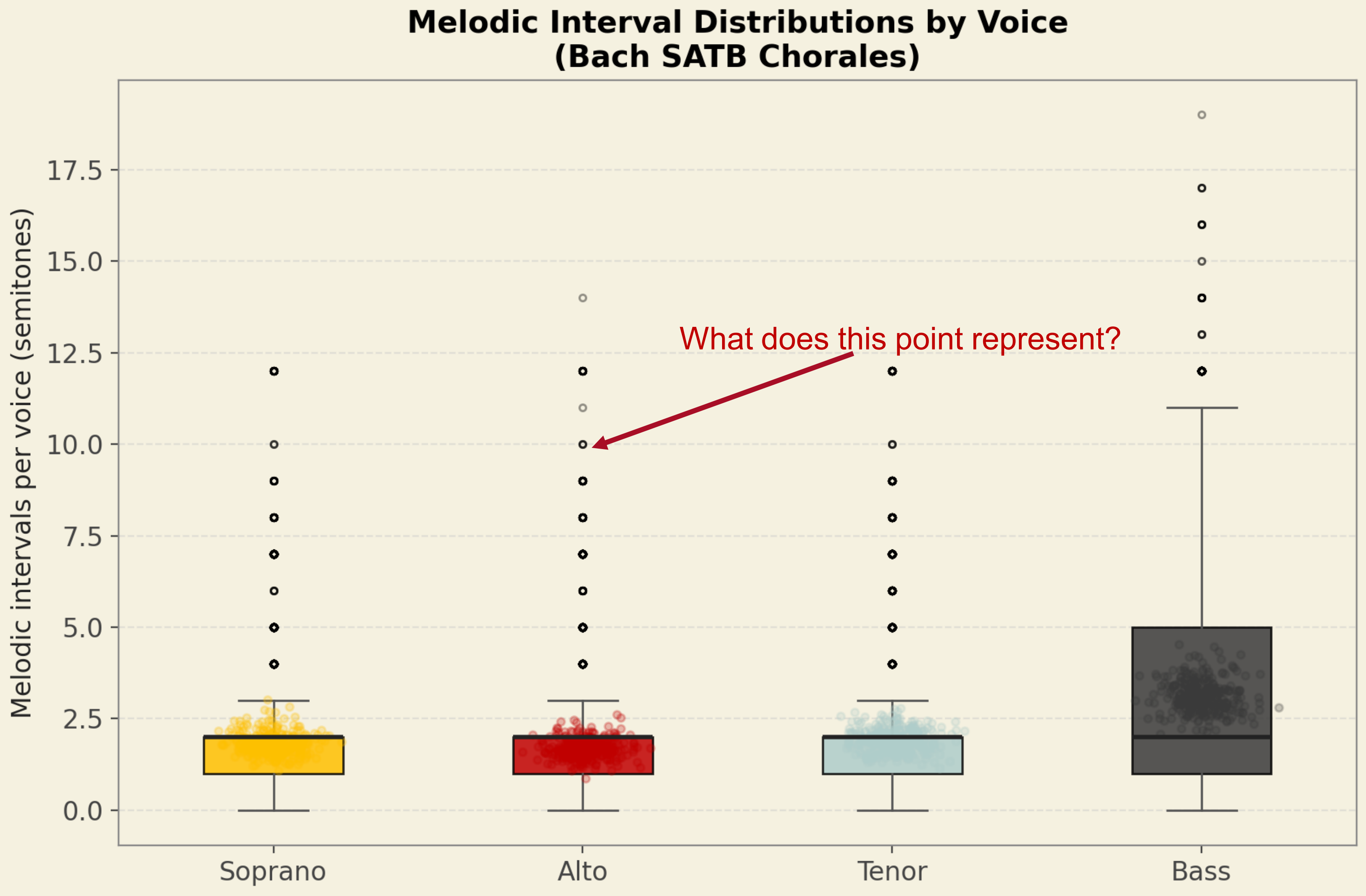


Quartiles and the Boxplot

- Quartiles divide the sorted data into four equal parts
- The box shows the middle 50% of the data (25^{th} to 75^{th} percentile)




Melodic Interval Distributions by Voice
(Bach SATB Chorales)
Mean melodic interval per chorale (semitones)
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
Soprano
Alto
Tenor
Bass
What does this point represent?


Melodic Interval Distributions by Voice
(Bach SATB Chorales)
Melodic intervals per voice (semitones)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Soprano
Alto
Tenor
Bass
What does this point represent?

Statistical tests

Moving from description to modeling

Our research project

Bass parts in Bach chorales are the most difficult to sing.

A corpus of 320 Bach chorales from music21

Large jumps (%), accidentals (%), tessitura extremes, syncopation

Statistical tests (t-test, Chi squared test, ANOVA)

Are the result statistically significant?

Bass do or do not have the short end of the stick

Statistical testing

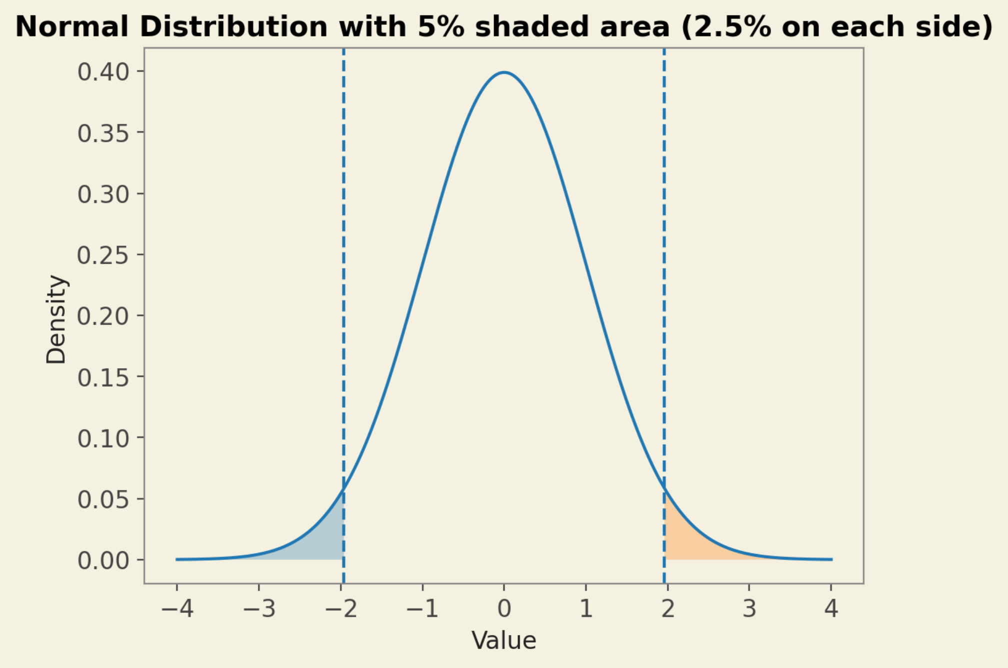
- Descriptive statistics and visualization are very helpful and sometimes show obvious differences
- However, could the differences we've seen be just due to chance?

Hypothesis testing

- Null hypothesis (H_0)
  - H_0: There is no difference between the groups
  - H_1: There is a genuine difference
- The p-value shows us the probability of observing data at least as extreme as observed, assuming the null hypothesis is true
- Conventionally, we reject H_0 if $p<0.05$ or 0.01

P-value

- 1% = 1/100 -> 0.01
- 5% = 5/100 -> 0.05



P-value misconceptions

- P-value is the probability that H_0 is true
- P=0.06 means there is no effect, p=0.04 means there is
- If p is smaller than 0.05 that must mean the H_0 is true (we only ~~failed~~ to reject H_0)
- A small p-value means there is a large effect

Comparing two groups: t-test

Concept

- Used to compare the means of two groups
- Assumes both groups are approximately normally distributed
- H_0: The two group means are equal ($\mu_1 = \mu_2$)
- The t-statistic measures how many standard errors apart the means are
- $t = (\bar{x}_1 - \bar{x}_2) / SE$
- Larger $|t| \rightarrow$ more evidence against H_0

Musicological example

- Do pianists have larger hands than people who do not play piano?
- Group 1: piano students from conservatory
- Group 2: Business students from university in the same city
- H_0: mean pianist hand span = mean business student hand span

Table 25: Comparison between hand spans of Adult Pianists and Business Students

		ADULT PIANIST DATABASE				BUSINESS STUDENT DATABASE					Test – diff. between means
		Arithmetic mean		Standard deviation			Arithmetic mean		Standard deviation		P value
	Sample	Inches	Cm	Inches	Cm	Sample	Inches	Cm	Inches	Cm	
All respondents 1-5 span*	473	8.2	21.0	0.71	1.81	216	8.0	20.3	0.72	1.82	0.0000
All respondents 2-5 span*	473	6.4	16.2	0.58	1.47	210	6.4	16.1	0.68	1.72	0.4371
Males 1-5 span	159	8.9	22.6	0.56	1.43	123	8.3	21.2	0.62	1.57	0.0000
Females 1-5 span	314	7.9	20.1	0.53	1.35	93	7.6	19.2	0.60	1.52	0.0000

R. Boyle, R. Boyle, and E. Booker, "Pianist Hand Spans: Gender and Ethnic Differences and Implications for Piano Playing," in 12th Australasian Piano Pedagogy Conference 2015, 8–29

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Do basses have larger melodic jumps?

4. Ach Gott und Herr

(Soll's ja so sein, Cantate No. 48. Ich elender Mensch. B.A. 10. S.288, BWV 48.3)

M. Rutilius (1551-1618) J.S. Bach (Orig. As hymnodus sacer, Leipzig 1625)

Do basses have larger melodic jumps?

```python
def get_melodic_intervals(midi_pitches):
    """
    Compute absolute melodic intervals (in semitones)
    between consecutive notes, ignoring rests
    """
    intervals = []
    for i in range(1, len(midi_pitches)):
        iv = abs(midi_pitches[i] - midi_pitches[i-1])
        intervals.append(iv)
    return intervals
```

Sopranoes: 1, 2, 2, 0, 2, 2, 1, 2, 2, 1, 2, 2, 1, 3, 2, 5, 1, 1, 2, 0, 2, 2, 2, 2, 7, 2, 2, 1, 2, 2, 2, 2, 2, 2, 0

Basses: 7, 5, 7, 10, 2, 1, 7, 1, 1, 7, 2, 2, 1, 2, 2, 7, 12, 5, 3, 2, 2, 7, 0, 7, 5, 7, 1, 2, 2, 1, 2, 2, 1, 2, 1, 1, 2, 1, 1, 2, 1, 2, 2, 12

Do basses have larger melodic jumps?